



University of Asia Pacific

Department of Computer Science and Engineering

CSE 437: Robotics Lab

LAB REPORT

Experiment Number: 02

Experiment Title: Design and Implementation of a Fire Fighting Robot.

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1. Experiment Name

Design and Implementation of a Fire Fighting Robot.

2. Objective

The main objective of this project is to design and implement a fire-fighting robot capable of detecting fire automatically and extinguishing it using a water pumping mechanism.

The robot is designed to:

- detect flame using a flame sensor,
- move towards the fire source,
- activate a water pump to extinguish the fire,
- and stop operation after the fire is removed.

This project helps to understand the practical application of robotics, sensors, motor control, and embedded systems in real-world safety and rescue operations.

3. Apparatus / Hardware & Software Requirements

Hardware:

- Microcontroller board
- Flame sensor module
- Motor driver module
- DC motors with wheels
- Water pump
- Water pipe and small water container
- Relay module (for pump control)
- Robot chassis
- Battery and power supply
- Jumper wires

Software:

- Arduino IDE (for writing and uploading the program)
- Embedded C / Arduino programming language
- Tinkercad for simulation

4. Circuit Diagram / Schematic

Flame Sensor Connections

Three flame sensors are used to detect the presence and direction of fire.

Sensor Position	Sensor Position
Left flame sensor	A0
Center flame sensor	A1
Right flame sensor	A2

DC Motor and Motor Driver (L293D) Connections

A dual H-bridge motor driver IC (L293D) is used to control two DC motors.

L293D Input Pin	Arduino UNO Pin	Description
IN1	D8	Left motor direction control
IN2	D9	Left motor direction control
IN3	D10	Right motor direction control
IN4	D11	Right motor direction control
EN1	5V	Enable pin for left motor
EN2	5V	Enable pin for right motor

DC Motor Connections

Motor	L298D Output Pins
Left DC motor	OUT1 and OUT2
Right DC motor	OUT3 and OUT4

Servo Motor Connection

The servo motor is used to control the direction of the water spraying nozzle.

Servo Pin	Arduino UNO Pin
Signal	D6

VCC	5V
GND	GND

Power supply connections

Component	Power Connection
Arduino UNO	USB power
L293D motor supply (Vcc2)	9V battery
Sensors and servo	Arduino 5V
All ground	Common ground

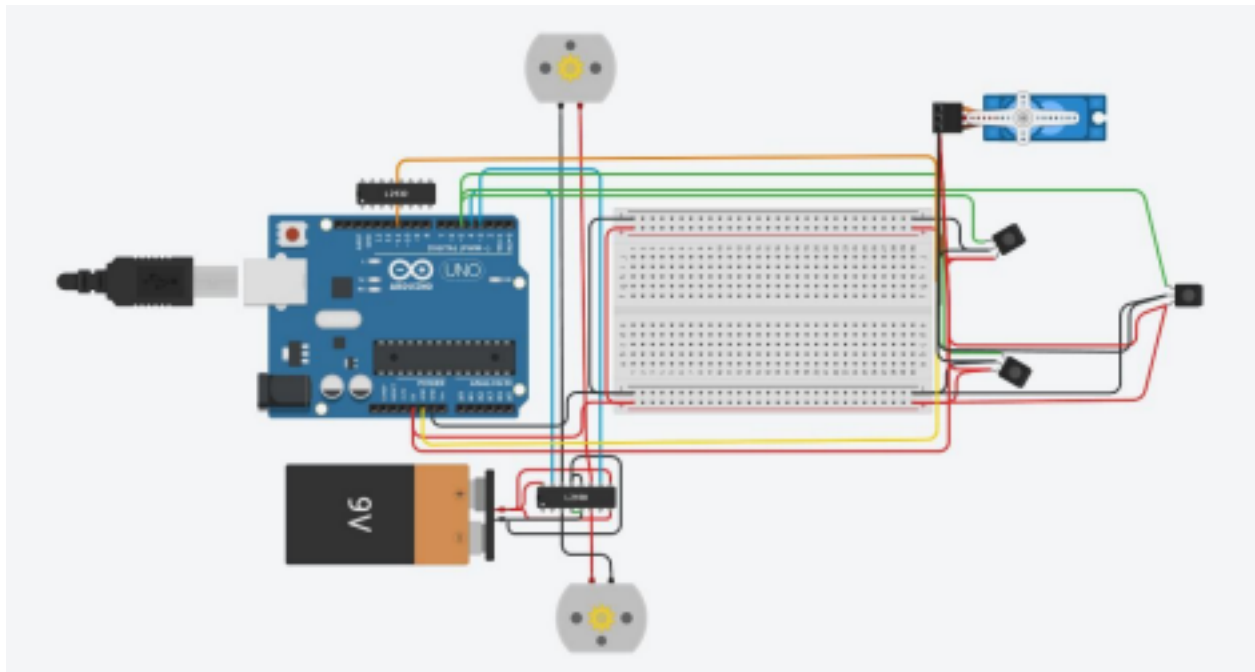


Fig 4.1: Circuit Diagram for Fire Fighting Robot

5. Code / Assembly Program

```
#include <Servo.h>
/* ----- Flame sensor pins ----- */
#define FLAME_LEFT A0
#define FLAME_CENTER A1
#define FLAME_RIGHT A2
/* ----- Motor driver pins ----- */
```

```
#define IN1 8
#define IN2 9
#define IN3 10
#define IN4 11
/* ----- Servo pin ----- */
```

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```
#define SERVO_PIN 6
```

```
Servo nozzle;
/* ----- Adjust this value after testing ----- */ int
flameThreshold = 500;
```

```
void setup()
{
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT); pinMode(IN3,
  OUTPUT); pinMode(IN4, OUTPUT);
```

```
  nozzle.attach(SERVO_PIN);
  nozzle.write(90); // center position
```

```
  Serial.begin(9600);
}
```

```
void loop() {
  int leftFlame = analogRead(FLAME_LEFT); int
  centerFlame = analogRead(FLAME_CENTER);
  int rightFlame = analogRead(FLAME_RIGHT);
```

```
  Serial.print(leftFlame);
  Serial.print(" ");
  Serial.print(centerFlame);
  Serial.print(" ");
  Serial.println(rightFlame);
```

```
  // Fire detected in front
  if (centerFlame < flameThreshold)
  {
    stopRobot();
    sprayWater();
  }
```

```
// Fire detected on left
else if (leftFlame < flameThreshold)
{
turnLeft();
}
```

```
// Fire detected on right
```

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```
else if (rightFlame < flameThreshold)
{
turnRight();
} // No fire
else
{
moveForward();
}
```

```
delay(100);
}
```

```
/* ----- Motor control ----- */
```

```
void moveForward()
{ digitalWrite(IN1,
HIGH);
digitalWrite(IN2, LOW);
```

```
digitalWrite(IN3, HIGH);
digitalWrite(IN4, LOW);
}
```

```
void turnLeft() {
digitalWrite(IN1, LOW);
digitalWrite(IN2, HIGH);
```

```
digitalWrite(IN3, HIGH);
digitalWrite(IN4, LOW);
}
```

```
void turnRight() {
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
```

```

digitalWrite(IN3, LOW);
digitalWrite(IN4, HIGH);
} void stopRobot() {
digitalWrite(IN1, LOW);
digitalWrite(IN2, LOW);

digitalWrite(IN3, LOW);
digitalWrite(IN4, LOW);

```

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```

}
/* ----- Servo movement (nozzle) ----- */

void sprayWater()
{
for (int pos = 70; pos <= 110; pos++)
{
nozzle.write(pos);
delay(15);
}
for (int pos = 110; pos >= 70; pos--)
{
nozzle.write(pos);
delay(15);
}
}

```

6. Output / Observations

During testing, the following observations were made:

- The flame sensor successfully detects the presence of fire within a short distance. •
When fire is detected, the robot moves toward the flame.
- After reaching the fire source, the water pump is automatically activated. •
Water is sprayed on the flame and the fire is extinguished.
- Once the flame is no longer detected, the pump is turned off and the robot stops. The

robot responds correctly to different fire positions by changing its movement direction. 7.

Result

The fire-fighting robot was successfully designed and implemented.

The robot can:

- detect fire,
- navigate toward the fire source,
- and extinguish the fire automatically.

All hardware components and control logic worked properly according to the project objectives.

8. Conclusion

This project demonstrates a simple and effective implementation of an autonomous fire-fighting robot using sensors and motor control. The system shows how robotics can be applied in

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safetycritical environments such as industrial plants, warehouses, and laboratories. Although the robot works well for small-scale fire detection and extinguishing, future improvements can include better navigation, longer sensing range, and the use of multiple sensors for higher accuracy. Overall, the project successfully fulfills its design and implementation goals.

